

Arjun Patel

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EDUCATION

Imperial College London | *MSc Computer Science (Distributed Systems)* 2018-2019

- Thesis on low-latency event stream processing: achieved 45ms end-to-end latency on Kafka clusters with 2.3M events/sec throughput
- Distinction grade (GPA 3.91/4.0); recipient of the Shell Technology Fellowship; led the Distributed Systems reading group with 32 members

University of Amsterdam | *BSc Computer Science (Algorithms & Data Systems)* 2015-2018

- Summa cum laude (GPA 3.87/4.0); published 1 peer-reviewed paper on time-series compression in streaming contexts
- President of the Data Science Club (200+ members); mentored 15 junior developers in systems design fundamentals

EXPERIENCE

Datadog | *Senior Data Engineer — Streaming Infrastructure* 2022-Present

- Architected and deployed real-time metric aggregation pipeline processing 8.2B events/day across 50+ Kafka clusters; reduced query latency from 2.8s to 340ms (88% improvement) for dashboard refreshes serving 15K+ customers
- Led migration of legacy batch ETL to Apache Flink-based stateful processing, cutting infrastructure costs by \$2.1M annually and improving metric freshness from 5min to <500ms for 97% of ingestion paths
- Designed and open-sourced custom Kafka consumer group rebalancer reducing rebalance time from 18s to 2.1s, preventing 4-hour monthly data freshness windows; adopted internally and by 12+ companies (800+ GitHub stars)

Stripe | *Data Engineer — Payment Streams* 2020-2022

- Built production streaming pipeline (Kafka + Flink) ingesting 650K transactions/sec with exactly-once semantics; detected and prevented \$47M in fraudulent transactions annually via real-time ML model serving with <120ms latency
- Optimized stream join operations across 8 billion daily payment events, reducing state store memory footprint by 62% (from 840GB to 319GB per node) and enabling 40% larger transaction volume capacity per cluster
- Mentored team of 6 junior engineers on distributed systems best practices; conducted 18 technical interviews and promoted 3 individuals to senior roles; drove 15+ architectural design reviews

Wise (TransferWise) | *Data Engineer — Real-Time Analytics* 2019-2020

- Engineered real-time settlement pipeline processing 125K cross-border transfers/hour with sub-second latency; improved settlement accuracy from 99.2% to 99.97% using custom fraud detection stream processors
- Reduced data warehouse refresh cycle from 6 hours to 4 minutes by migrating CDC pipeline from Logstash to Kafka Connect with Debezium, unlocking 24/7 real-time reporting for Finance and Risk teams
- Implemented comprehensive monitoring and alerting for 18 critical data pipelines using Prometheus and custom Kafka metric exporters; achieved 99.98% uptime SLA and reduced MTTR from 35min to 8min

Thoughtworks | *Software Engineer — Data Infrastructure* 2017-2019

- Developed internal streaming data platform using Apache Kafka and Apache Spark; onboarded 12 cross-functional teams and reduced data pipeline development time from 3 weeks to 3 days via reusable abstractions
- Designed schema evolution strategy for event streams supporting 30+ event types with backward/forward compatibility; reduced production incidents related to schema mismatches by

100% (from 8/quarter to 0)

- Authored technical blog posts and internal documentation on stream processing patterns; reached 45K+ readers and contributed to 4 tech talks at company conferences

PROJECTS

FlowSync | *Kotlin, Kafka, Apache Flink, RocksDB, Docker* 2021-2023

- Open-source framework for building exactly-once streaming pipelines with automatic state management and checkpointing; 2,100+ GitHub stars, 180+ forks, 45K downloads/month on Maven Central
- Benchmarked throughput at 1.8M events/sec per node with sub-100ms latency tail percentiles (p99); reduced boilerplate code for new pipeline development by 70% compared to raw Flink APIs

RiverTelemetry | *Go, Prometheus, ClickHouse, TimescaleDB* 2022-Present

- High-performance metrics collection agent for distributed systems; 650+ GitHub stars, used in production by 200+ companies to monitor 50K+ services globally
- Achieves <5% CPU overhead with lock-free ring buffers; sends telemetry at 100K metrics/sec with <50ms latency; reduced disk I/O by 85% vs. competing solutions (Telegraf, collectd)

PipelineSync | *Python, SQLAlchemy, PyArrow, Airflow, S3* 2019-2021

- Data pipeline testing and validation framework supporting incremental validation; 920+ GitHub stars, adopted by 35+ organizations; reduces testing time from 2 hours to 8 minutes for 100GB datasets
- Integrates schema validation, row-level hashing, and statistical drift detection; caught 99.2% of data quality regressions in production pipelines when applied as pre-deployment checks

EXTRACURRICULAR

Apache Kafka Community | *Contributor & KIP Review Committee Member* 2020-Present

- Contributed 12 pull requests merged into core Kafka codebase, including KIP-694 (transaction coordinator improvements); reviews averaged 3.2 PRs/week across 8+ KIPs affecting millions of users
- Co-authored technical RFC for stream recovery optimization adopted by Confluent engineering; improvements reduce bootstrap time by 40% for 10TB+ clusters

Data Engineering Europe Conference | *Speaker & Workshop Organizer* 2021-2023

- Delivered 3 keynote talks on 'Streaming at Scale' reaching 1,200+ attendees at conferences in Berlin, Amsterdam, London; talks averaged 4.8/5.0 rating with 80K+ YouTube views combined
- Designed and led 2-day hands-on workshop on production Kafka architectures; mentored 60+ engineers; 92% completion rate with post-workshop survey showing 8.9/10 satisfaction score

TECHNICAL SKILLS

Languages – Python, Kotlin, Go, SQL, Scala, Java

Technologies – Apache Kafka, Apache Flink, Spark Streaming, Apache Beam, Debezium, Airflow, ClickHouse, PostgreSQL, Redis, AWS (S3, Kinesis, EC2), GCP (BigQuery, Dataflow)

Tools – Docker, Kubernetes, Terraform, Git, Prometheus, Grafana, DataDog, Jira, Confluence